

Visual Explainer: Angular Manifold Routing

Step 1: Embed Tokens

Text inputs are converted into numerical vectors called embeddings.

Step 2: Normalize

Vectors are normalized so they lie on a unit hypersphere. This means only direction matters.

Step 3: Partition the Sphere

The hypersphere is divided into angular sectors that serve as routing buckets.

Step 4: Route by Direction

Each token is assigned to the sector closest to its direction.

Step 5: Sparse Compute

Only sectors containing tokens activate compute modules.

Key Insight

Tokens naturally cluster into fewer sectors than expected. Even if thousands of routing sectors exist, only a fraction are active at any time.